Data Lake Implementation using AWS Glue and Athena

Services: S3, Glue, Athena, IAM

Description: Create a data lake on Amazon S3, catalog the data using AWS Glue, and perform SQL queries using Amazon Athena. Configure IAM policies for secure access.

Skills: Data cataloging, serverless querying, data lake architecture.

Definition: The Data Lake Implementation using Amazon S3, AWS Glue and Amazon Athena provides a serverless architecture for storing, cataloging, and querying structured data.Raw data (CSV files) is stored in S3, automatically discovered using an AWS Glue crawler, converted into metadata tables in the Glue Data Catalog, and queried using SQL through Athena. Access is controlled using AWS Identity and Access Management policies.

Scope of the Project:

The scope of this project is to design and implement a basic data lake architecture using AWS managed services.

This project covers:

- **Data Storage in S3**
  Stores raw structured data files (CSV) inside an S3 bucket.
- **Data Cataloging using Glue**
  Automatically scans and identifies schema using a Glue crawler.
- **Serverless Querying with Athena**
  Executes SQL queries on the data without managing servers.
- **Secure Access Control using IAM**
  Controls permissions for Glue and Athena access.

Methodology

1. S3 + Glue + Athena (Best Method)

- Uses all required AWS services.
- Fully serverless architecture.
- No database server provisioning required.
- Free-tier conscious with small datasets.
- Easy to implement and scalable.
- Suitable for analytics and reporting workloads.

2. Manual S3 + Athena External Table

- Upload data to S3.
- Manually create Athena external table using SQL.
- Query data directly without Glue crawler.
- Simpler but less automated.
- Useful for very small static datasets.

AWS Services used in the Project

1. Amazon S3

Definition:
Amazon S3 is an object storage service for storing and retrieving data.

Purpose:
Stores raw data files and Athena query results.

Role:

- Stores CSV dataset
- Stores Athena output results
- Serves as data lake storage layer

2. AWS Glue

Definition:
AWS Glue is a serverless data integration service for discovering and cataloging data.

Purpose:
Automatically identifies data schema.

Role:

- Creates Glue database
- Runs crawler
- Creates metadata table in Data Catalog

3. Amazon Athena

Definition:
Amazon Athena is a serverless interactive query service.

Purpose:
Allows SQL analysis directly on S3 data.

Role:

- Queries Glue catalog tables
- Performs aggregation and analysis
- Stores query results in S3

4. AWS IAM

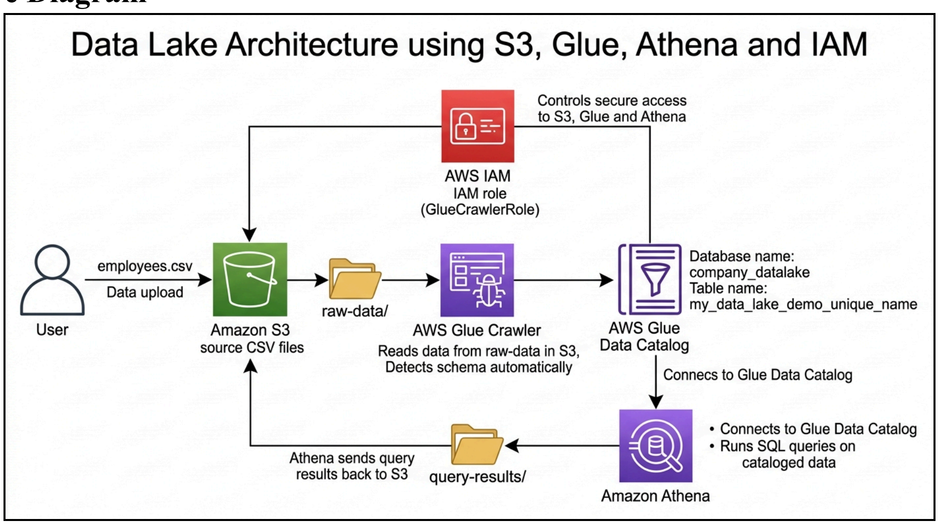
Definition:
AWS Identity and Access Management manages permissions securely.

Purpose:
Controls access to Glue, Athena and S3 resources.

Role:

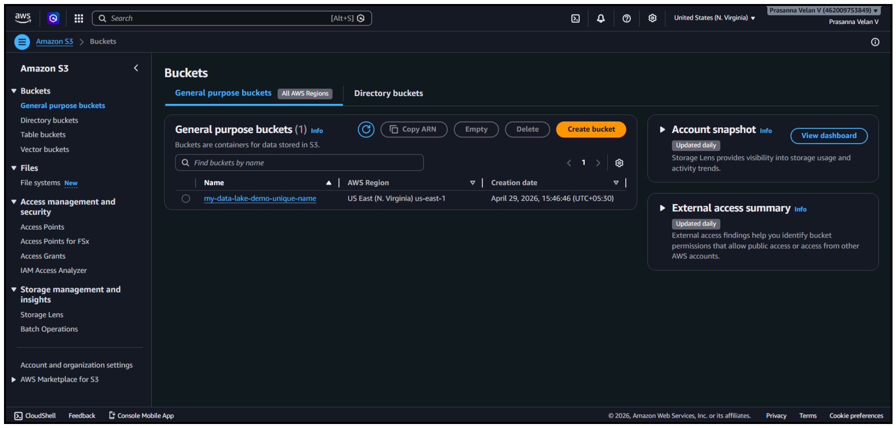
- Glue service role permissions
- S3 access permissions
- Secure data access

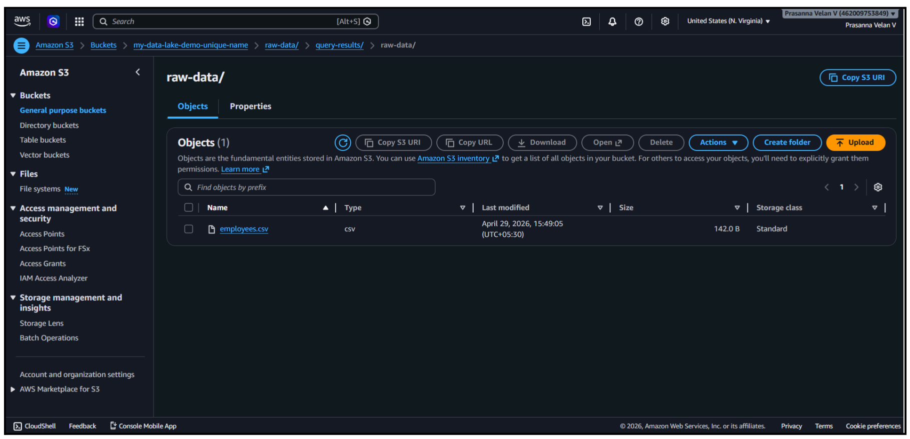
Architecture Diagram



Implementation Steps to Do the Project:
Step 1: Create S3 Bucket

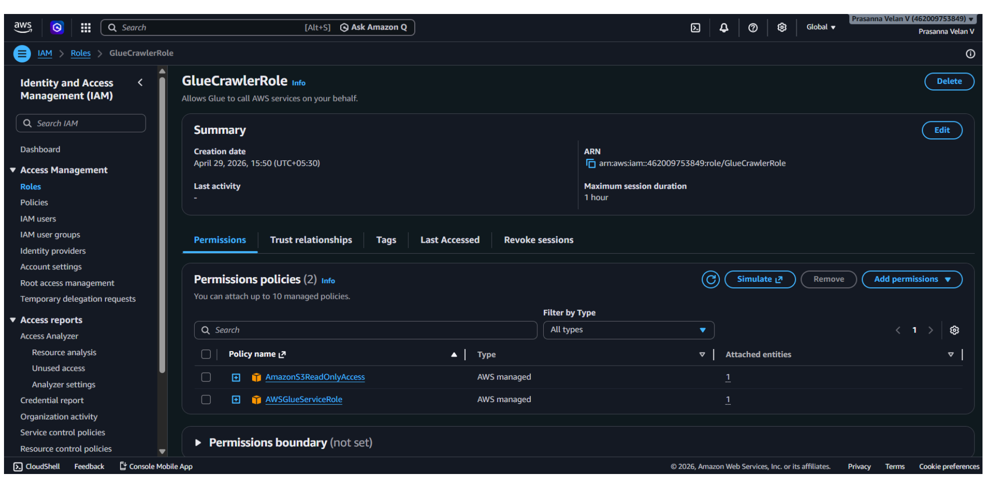
- Go to AWS Console → S3 → Create Bucket.
- Enter bucket name: **my-data-lake-demo**
- Create the bucket.
- Inside the bucket, create folders: raw-data/ and query-results/
- Upload the dataset file **employees.csv** into the raw-data/ folder.
- Verify that the file is successfully uploaded in the S3 bucket.





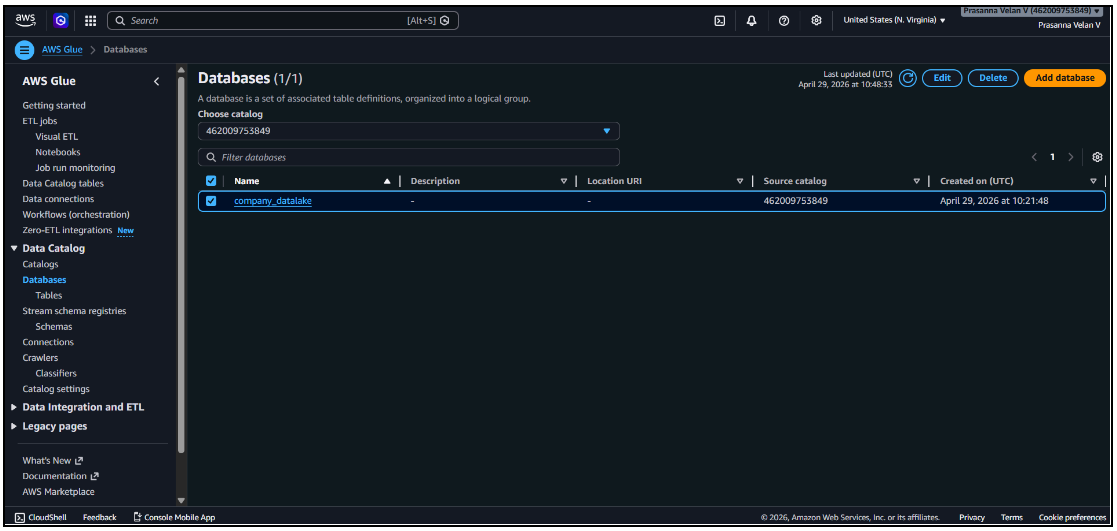
Step 2: Create IAM Role for Glue

- Go to IAM → Roles → Create Role.
- Select trusted entity as **AWS Service**.
- Choose use case as **Glue**.
- Attach policies: AWSGlueServiceRole ,AmazonS3ReadOnlyAccess
- Name the role: **GlueCrawlerRole** and Review and create the role.



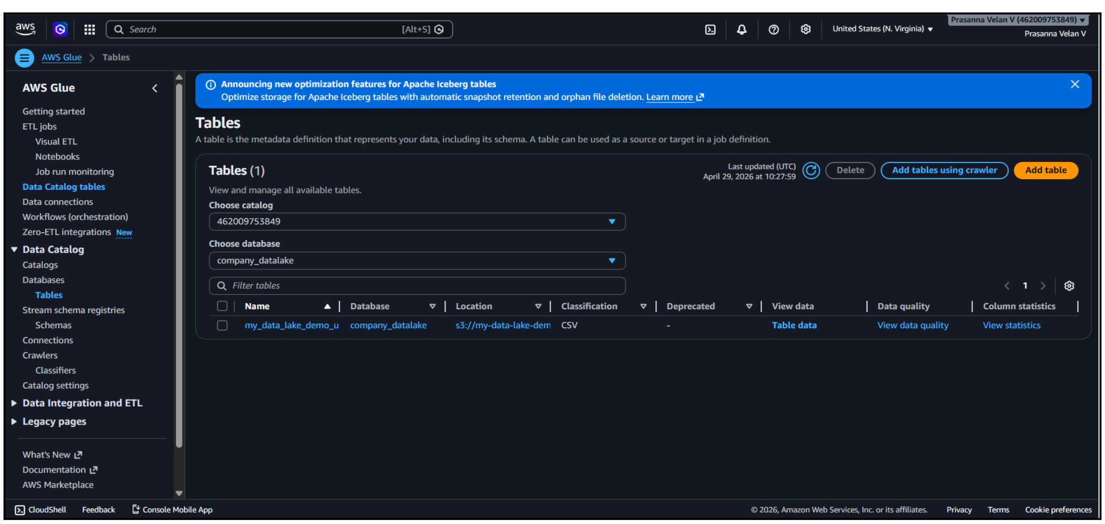
Step 3: Create Glue Database

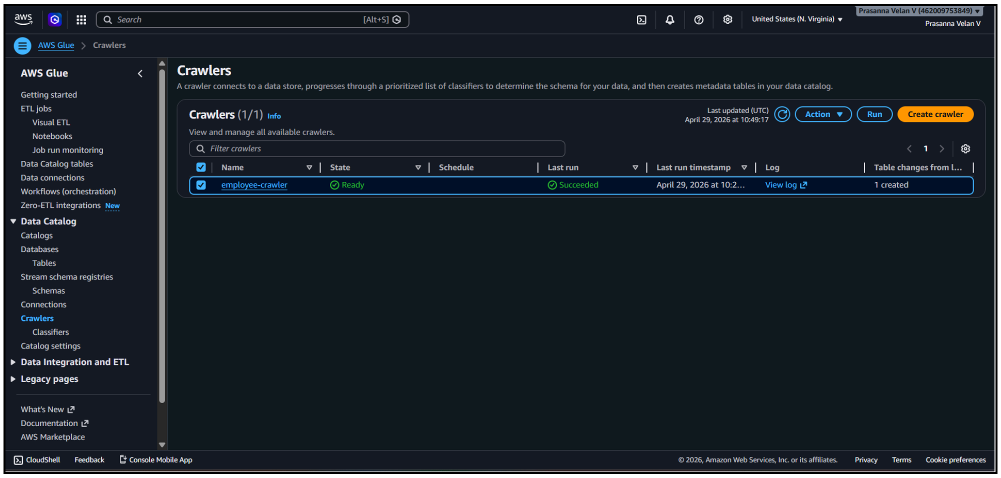
- Go to AWS Glue → Data Catalog → Databases.
- Click **Create Database**.
- Enter database name: **company_datalake**
- Save the database.



Step 4: Create Glue Crawler

- Go to AWS Glue → Crawlers → Create Crawler.
- Enter crawler name: **employee-crawler**
- Select data source as Amazon S3.
- Enter S3 path:
- s3://my-data-lake-demo-unique-name/raw-data/

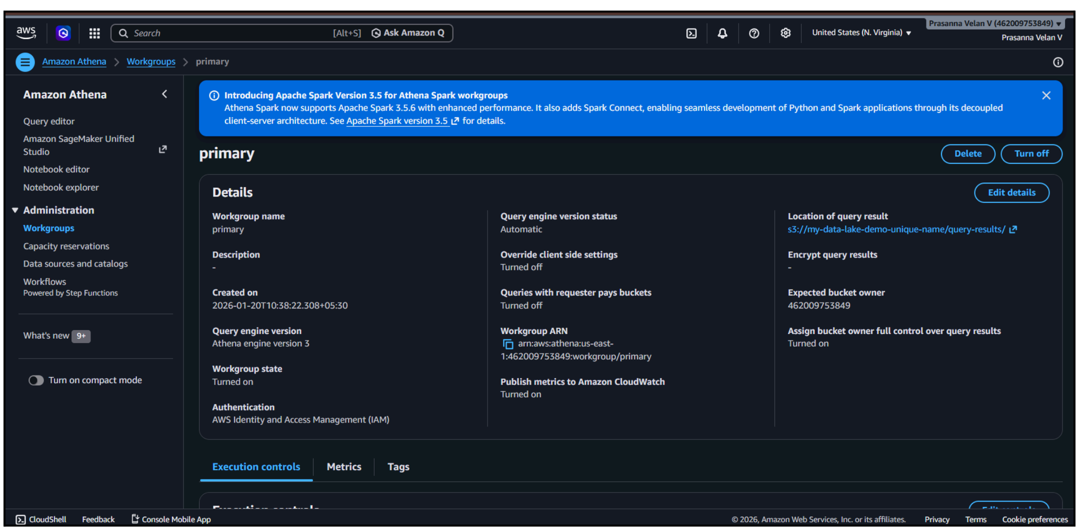




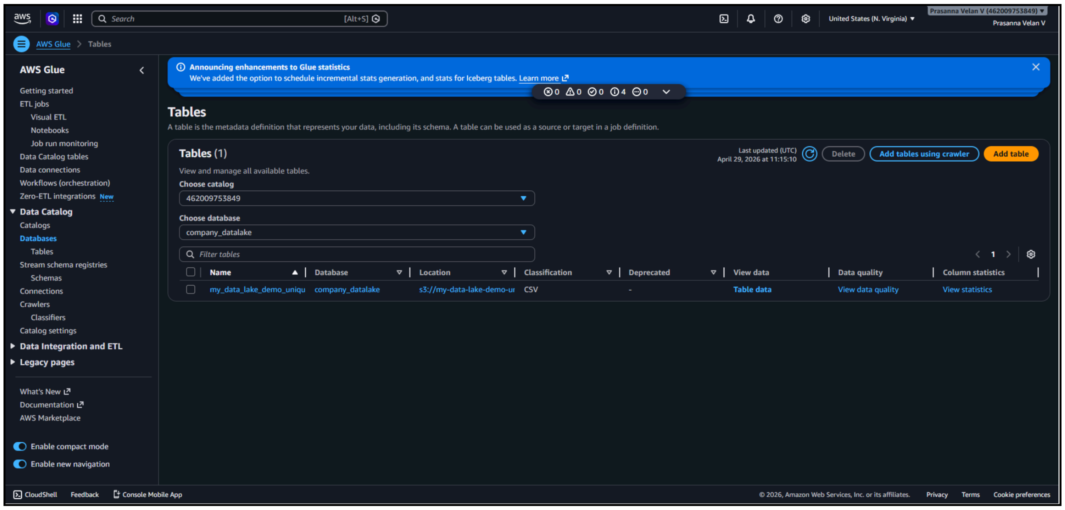
- Select IAM role: **GlueCrawlerRole**
- Choose target database: **company_datalake**
- Complete the crawler configuration.
- Run the crawler.
- Verify the crawler creates the table:
- My_data_lake_demo_unique_name
- Check detected schema and column structure.

Step 5: Configure Athena

- Go to Amazon Athena → Query Editor.
- Configure query result location:
- s3://my-data-lake-demo-unique-name/query-results/



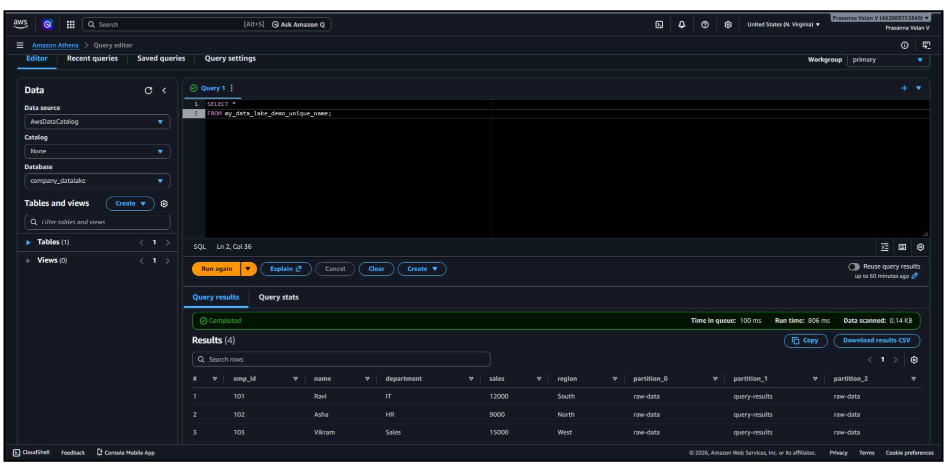
- Save settings.
- Select database:
- company_datalake



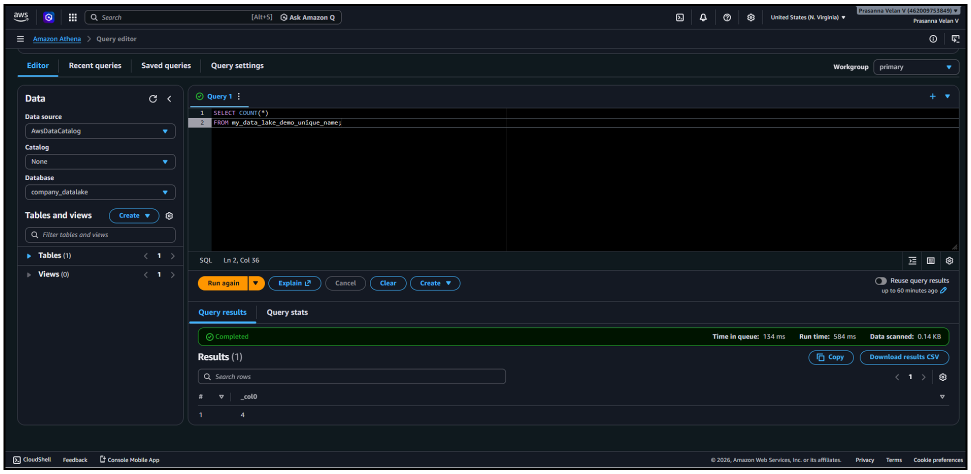
- Confirm table is visible in Athena.

Step 6: Run Athena Queries

- Execute test query to retrieve data:
- SELECT *
- FROM my_data_lake_demo_unique_name;
- Execute row count query:
- SELECT COUNT(*)
- FROM my_data_lake_demo_unique_name;



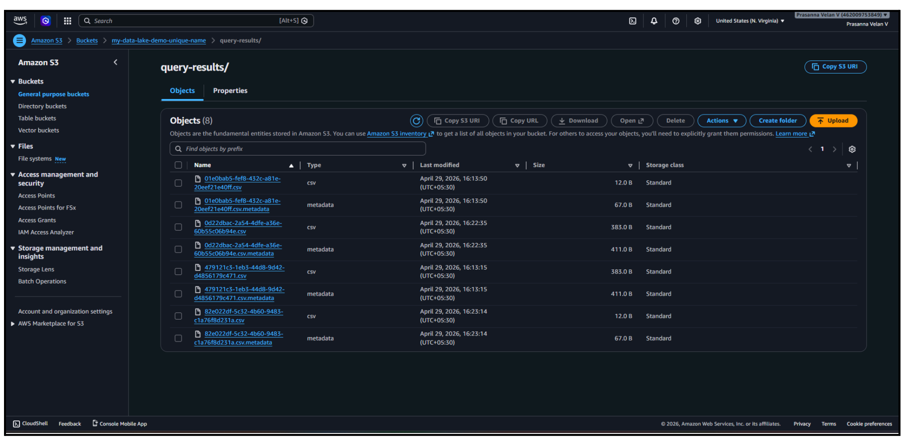
- Execute aggregation query:
- SELECT department,
- SUM(sales) and FROM my_data_lake_demo_unique_name and GROUP BY department



- Verify query output is generated successfully.

Step 7: Verify Query Results

- Review query results in Athena output pane.
- Go to S3 bucket → query-results/ folder.
- Confirm Athena stores result files automatically.
- Validate returned records and aggregation results.



Troubleshooting

Issue 1: Athena "Table Not Found"

Athena returned error:TABLE_NOT_FOUND

The query referenced an incorrect table name.

Solution: Use the actual Glue crawler-generated table: my_data_lake_demo_unique_name

instead of: Employees -Re-run the query using the correct table.

Issue 2: Athena Query Output Location Not Set

Athena returned error: No output location provided =Athena requires an S3 location to store query results.

Solution: Configure query result location: s3://my-data-lake-demo-unique-name/query-results/

Save settings and run query again.

Issue 3: Access Denied in Glue Crawler

The Glue crawler failed due to insufficient S3 permissions.

Solution:

- Attach policy: AmazonS3ReadOnlyAccess
- to the Glue IAM role: GlueCrawlerRole
- Re-run the crawler.

Conclusion

The Data Lake Implementation using AWS Glue and Athena provides a simple, scalable and serverless analytics solution for storing and querying structured data. By integrating Amazon S3, AWS Glue, Amazon Athena and IAM, the project demonstrates core data lake architecture concepts including storage, cataloging, querying and secure access control.

The solution is cost-effective, free-tier conscious for small datasets, and suitable for learning foundational data engineering and analytics concepts.